Optimizing Job Search in the ABM: Summary of the Benchmark Exercise

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1 Overview

In the baseline ABM, which extends the data-driven network model of del Rio-Chanona et al. (2021), unemployed workers draw application effort from an empirical distribution, so search behaviour is imposed rather than chosen. We replace the effort draw with an optimizing rule and leave the rest of the model unchanged. We then compare behavioural and non-behavioural versions of that rule inside the ABM against the baseline, holding its calibrated parameters fixed. Table 1 lists each component and its closest counterpart in the literature.

Component	Our specification	Closest literature	Main difference
Application choice (all optimizing rules)	Rank found vacancies by surplus; apply while (chance of first success) × surplus $\geq \kappa w$	Simultaneous search (Chade and Smith, 2006); search intensity (Mortensen, 1977; Pissarides, 2000)	Discrete, myopic within the month, linear cost; exact optimal portfolio only with equal success probabilities
Reservation wage	Empirical schedule declining with duration (all rules except the benchmarks)	Non-stationary search (van den Berg, 1990); evidence in Krueger and Mueller (2016)	Not derived from the model
Behavioural rules			
Sticky beliefs (main)	Perceived success erodes with own failures, decreasing-gain update	Optimistic, slowly revised beliefs (Spinnewijn, 2015; Mueller et al., 2021); decreasing-gain learning (Evans and Honkapohja, 2001)	This is our own designed rule, as in Appendix A
Bayesian beliefs (robustness)	Beta-Bernoulli update, one failure per month	Learning from unsuccessful search (Burdett and Vishwanath, 1988; Gonzalez and Shi, 2010)	Static application rule; learning does not move the reservation wage
Non-behavioural benchmarks (no reservation wage)			
Fixed beliefs	Constant perceived success probability	Stationary search intensity (Pissarides, 2000)	No equilibrium matching function
Full information	Perceived success = each occupation's realized hiring probability last month	Directed search with multiple applications (Albrecht et al., 2006; Galenianos and Kircher, 2009; Kircher, 2009); competitive search (Moen, 1997); urn-ball hiring (Burdett et al., 2001)	Adaptive (last month's values), not equilibrium probabilities

Table 1: The proposed decision rules and their closest counterparts in the literature.

2 Calibration strategy

In the baseline ABM, unemployed workers draw application effort from an empirical CPS schedule. Here we replace that draw with an optimizing rule and keep the rest of the model, including all calibrated parameters (d_u, γ_u, θ), unchanged. We pin down the new parameters of the decision rule outside the ABM, from micro evidence on job search, instead of adding them to the macro calibration on the unemployment and vacancy rates. There are three reasons. First, the unemployed rule deliberately ignores market tightness, so the two aggregate series carry almost no information about the parameters that shape effort over an unemployment spell. Those parameters would be weakly identified and would trade off against d_u and γ_u. Second, setting parameters with a direct micro counterpart from micro data, and leaving the remaining parameters to the macro calibration, is standard practice. It also lets us compare the decision rules holding the calibrated economy fixed. Third, the CPS effort profile used as an input in the baseline becomes an outcome of the optimizing

model. Its model counterpart can then be checked rather than imposed, and effort responds when the environment changes. The belief and cost parameters were chosen in a stand-alone simulation of individual job seekers to reproduce the shape and level of the CPS effort-by-duration profile, with reference to evidence on job seekers' beliefs (Mueller et al., 2021). This was an exploratory choice, not a formal estimation.

3 Behavioural decision rules

All rules apply to unemployed workers only. Employed workers keep the baseline on-the-job search rule.

Common structure. Each month, an unemployed worker b with previous wage w_b and unemployment duration τ samples up to 30 vacancies from their occupational neighbourhood, as in the baseline. Vacancies paying at least the empirical reservation wage $R_{b,\tau} = w_b\,\hat{r}(\tau)$, where $\hat{r}(\tau)$ is the estimated reservation-wage ratio by duration, form the candidate set $\mathcal{V}_{b,t}$. Occupational similarity enters as $\tilde{\rho}_{ij} \in (0, 1]$, the within-row percentile of the transition weight from the worker's occupation i to occupation j (own occupation = 1). The surplus of vacancy a is

$$\Delta V_a = (w_a - R_{b,\tau})\,\tilde{\rho}_{ij(a)}, \qquad a \in \mathcal{V}_{b,t}, \tag{1}$$

and candidates are ranked so that $\Delta V_1 \geq \Delta V_2 \geq \ldots$. With a perceived per-application success probability $p_{b,t}$ that is common across applications, the marginal benefit of the a-th application and the optimal number of applications are

$$MB_a = (1 - p_{b,t})^{a-1}\,p_{b,t}\,\Delta V_a, \qquad A^*_{b,t} = \max\{a \leq \bar{A} : MB_a \geq \kappa\,w_b\}, \qquad A^*_{b,t} \geq 1. \tag{2}$$

The cost is proportional to the worker's wage, so A^* does not depend on the wage level. The ceiling $\bar{A} = 30$ is the number of vacancies sampled. The floor of one application per month avoids an absorbing state of zero search and can be read as a UI work-search requirement. The rules differ only in how $p_{b,t} = \beta_{b,\tau}\,\bar{p}$ evolves. Beliefs are updated only on the worker's own failures, with no signal of market tightness.

Sticky beliefs (main specification). This is equation (10) of the formalization, with no positive signals while unemployed ($h = 0$):

$$\beta_\tau = \Pi_{[0,1]}\big(\beta_{\tau-1} - r\,e^{-\omega\tau}\big) \implies \beta_\tau = \max\Big\{0,\ \beta_0 - r\textstyle\sum_{s=1}^{\tau} e^{-\omega s}\Big\}, \tag{3}$$

so beliefs erode at a decaying rate towards the floor $\beta_\infty = \beta_0 - r\,e^{-\omega}/(1 - e^{-\omega})$.

Bayesian beliefs (robustness). The worker holds a Beta prior over per-application success with mean $\bar{p}$ and strength N_0, and counts one failure per month unemployed:

$$\beta_\tau = \frac{N_0}{N_0 + \tau}. \tag{4}$$

A variant that counts one failure per application was also tested. It could not reproduce the decline in effort at long durations (fewer applications mean fewer failures, so learning stalls) and was dropped.

Parameter	Sticky	Bayesian	Basis
β_0	1	1 (implied)	Normalization: full confidence at the start of a spell
$\bar{p}$	0.20	0.30	Chosen to reproduce the CPS effort profile; no direct empirical counterpart
ω	0.10	–	As in the formalization's comparative statics
r	0.10	–	Chosen for the decline after month 12. Implies $\beta_\infty \approx 0.05$
N_0	–	2	Best fit. A weak prior, worth two months of evidence
κ	0.015	0.015	Chosen to match the level of CPS effort
$\bar{A}$, $a_{\min}$	30, 1	30, 1	Vacancies sampled; work-search requirement
$R_{b,\tau}$	empirical	empirical	Baseline reservation-wage regression

Table 2: Behavioural rules: parameter values. A sticky variant with $\bar{p} = 0.3$, $r = 0.105$ fits the CPS profile best in the stand-alone simulation, but sets $\beta_\infty \approx 0$, so beliefs erode completely. In the ABM it performs similarly to the main specification and is not reported.

How realistic are these values? Two features need flagging. First, $\kappa = 0.015$ equals about 31 hours of wages per application ($0.015 \times 2{,}080$ hours), far above measured time spent on job search (Krueger and Mueller, 2010). Most of the cost must therefore be non-pecuniary (effort, stress, rejection). Second, the best-fitting belief paths erode almost completely: under the sticky rule, β falls by about 95% within three years, and the Bayesian prior is very weak. Mueller et al. (2021) instead find that job seekers' beliefs are optimistic and barely revised downward. Under both rules the decline in effort therefore comes from more belief erosion than the survey evidence suggests. However, this is calibrated to the CPS data, so our decision on this issue will depend on which source of information we wish to follow.

4 Non-behavioural benchmarks

These are the counterparts of the baseline non-behavioural model. They use the same optimizing rule but switch off both behavioural channels. There is no reservation wage, so every sampled vacancy is a candidate and the surplus is measured from zero, $\Delta V_a = w_a \, \tilde{\rho}_{ij(a)}$. Neither rule depends on unemployment duration.

Fixed beliefs. Beliefs never erode ($r = 0$ in the sticky rule): $p_{b,t} = \bar{p}$, with MB_a and A^* as in equation (2).

Full information. Perceived success equals each occupation's realized chance of being hired in the previous month,

$$p_a = \pi_{j(a),t-1}, \qquad \pi_{j,t} = \frac{1}{|\mathcal{V}_{j,t}|} \sum_{v \in \mathcal{V}_{j,t}} \frac{1}{n_{v,t}}, \tag{5}$$

where $n_{v,t}$ is the number of applicants to vacancy v. In the ABM each vacancy picks one applicant, so $\pi_{j,t}$ is the average probability of being picked in occupation j. Candidates are ranked by $p_a \Delta V_a$, and the worker applies while

$$MB_a = \Big[\textstyle\prod_{k<a}(1 - p_k)\Big] \, p_a \, \Delta V_a \;\geq\; \kappa \, w_b. \tag{6}$$

This is the only rule in which unemployed workers observe market tightness. It is a deliberate contrast to subjective beliefs.

Parameter	Fixed beliefs	Full information	Basis
$\bar{p}$	0.20	– (fallback only)	Sticky rule's belief at the start of a spell
κ	0.043	0.0093	Set so mean applications per seeker equal the baseline's (5.35); refined with short ABM runs
Reservation wage	none	none	Behavioural channel switched off

Table 3: Non-behavioural benchmarks: parameter values. Without a reservation wage the surplus is the whole wage, so the same cost would buy far more search effort than in the behavioural rules. Setting κ to match average effort puts all rules on the same footing: each searches about as much as the data imply. The full-information benchmark is close to a tipping point: a slightly higher cost lowers effort, raises unemployment and lowers every application's success probability, which can push the economy into a persistent high-unemployment state.

5 Results

All scenarios run the reference model (behavioural, cyclical on-the-job search, reservation wages) with its calibrated parameters held fixed, 10 simulations each, 2001–2024. Only the unemployed search rule changes.

	UER RMSE 2001–2019	LTUER RMSE 2001–2019	Mean UER 2001–2024	Applications per seeker
Baseline (CPS effort schedule)	0.0058 (0.0006)	0.101 (0.004)	0.056	5.35
Sticky beliefs	0.0068 (0.0004)	0.080 (0.005)	0.055	5.66
Bayesian beliefs (per month)	0.0067 (0.0005)	0.081 (0.006)	0.055	5.50
Fixed beliefs	0.0090 (0.0006)	0.096 (0.004)	0.064	5.51
Full information	0.0142 (0.0010)	0.101 (0.004)	0.068	5.40

Table 4: Fit to observed series. RMSE is computed per simulation and averaged across the 10 simulations, with the standard deviation across simulations in parentheses. The vacancy rate is exogenous and fits equally well in all scenarios (RMSE 0.0024–0.0030).

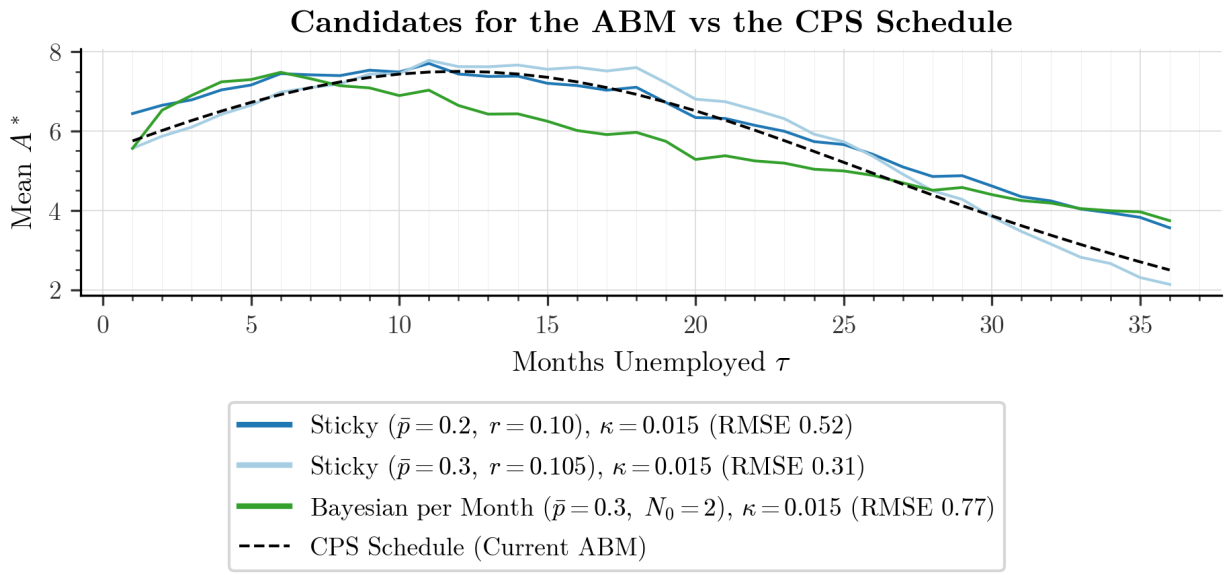


Figure 1: The decision rule outside the ABM: mean optimal applications by unemployment duration for simulated job seekers who are never hired, drawing from the initial vacancy stock. RMSE is measured against the CPS schedule that the baseline uses as input.

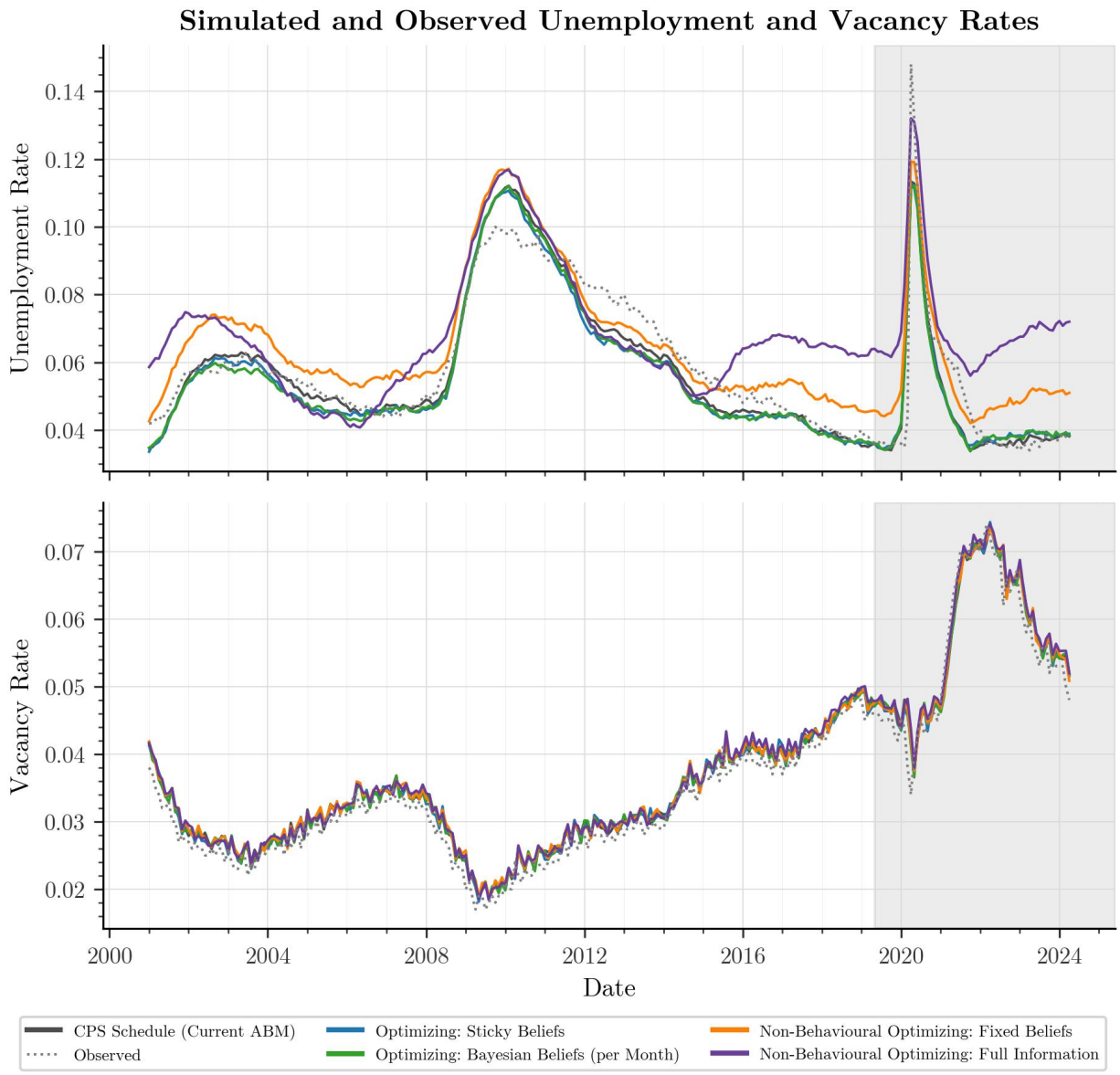


Figure 2: Unemployment and vacancy rates. The behavioural optimizing rules track the baseline. The non-behavioural benchmarks drift to persistently higher unemployment, even though they search as much as the baseline on average. The shaded area is outside the calibration window.

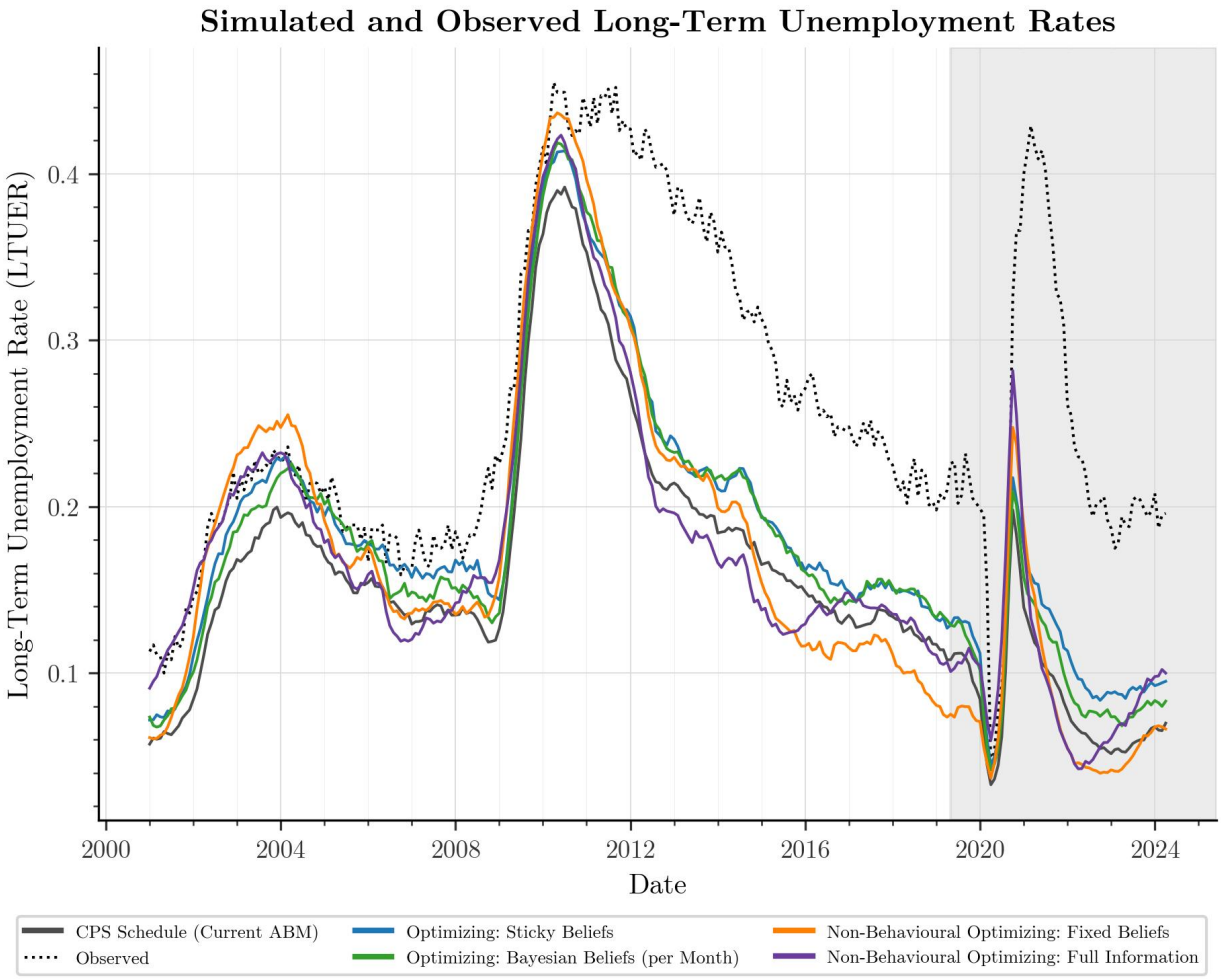


Figure 3: Long-term unemployment rate, simulated and observed.

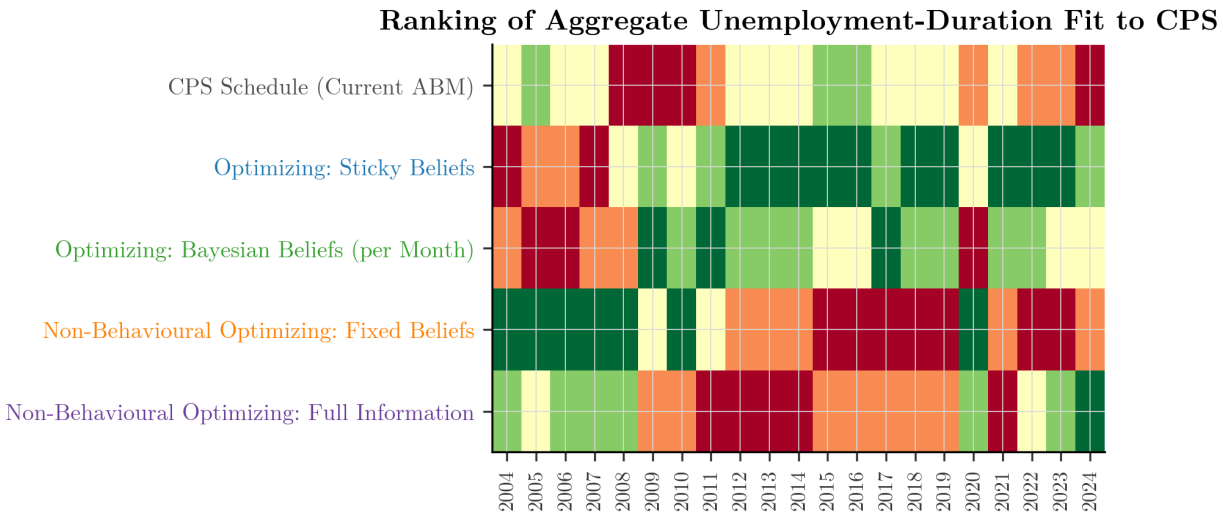


Figure 4: Annual ranking of scenarios by the Wasserstein distance between the simulated and CPS unemployment-duration distributions (green = closest).

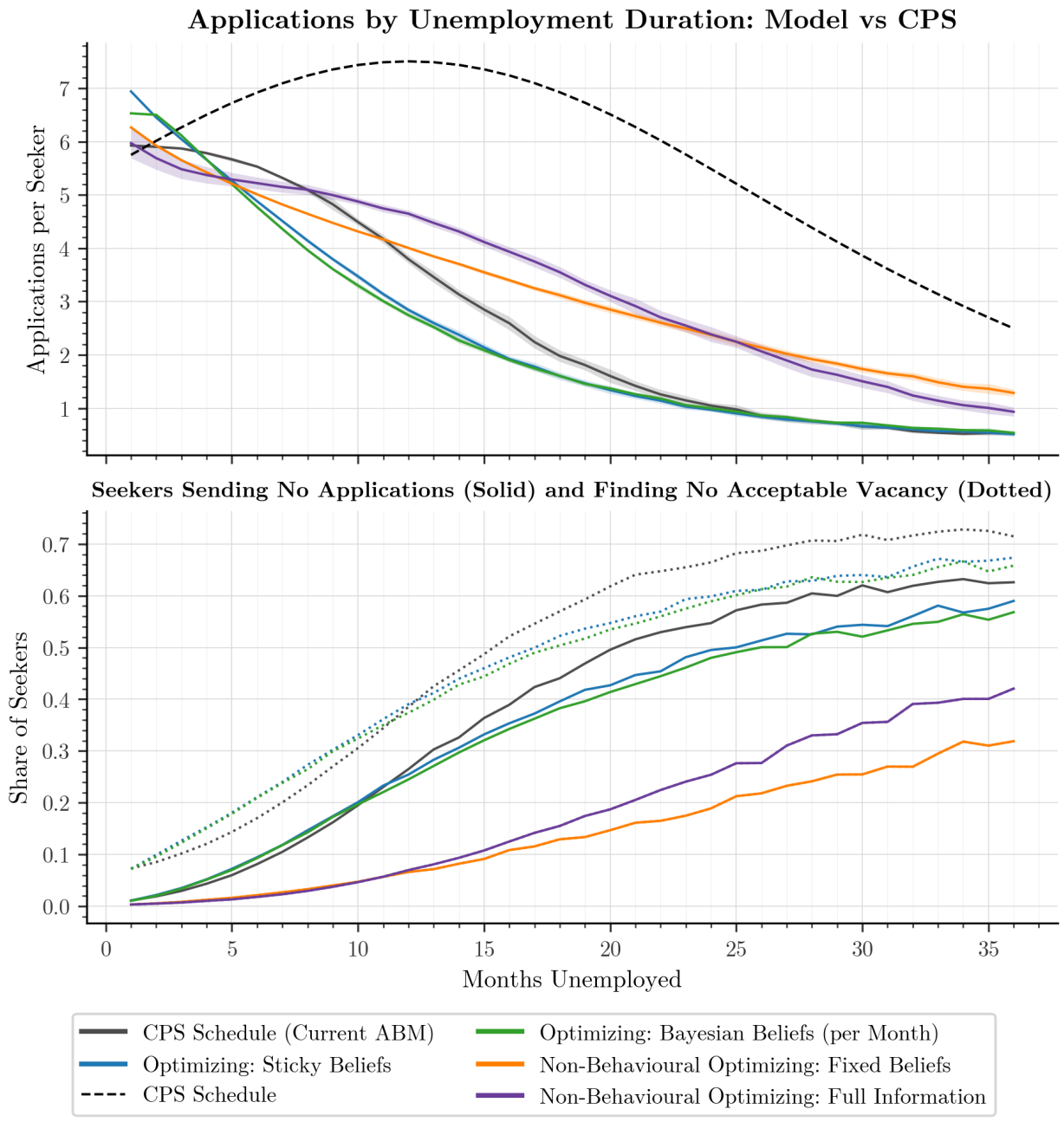


Figure 5: Realized applications by unemployment duration inside the ABM (top) and the share of seekers sending no applications or finding no acceptable vacancy (bottom). Realized effort declines with duration under every rule, including the baseline that draws effort from the CPS schedule. Long-duration seekers increasingly find no vacancy above their reservation wage among the vacancies they can reach.

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